

Inner Product space

$\mathbb{R}^2$ length $(x, y) \in \mathbb{R}^2$
 $|(x, y)| = \sqrt{x^2 + y^2}$

$\mathbb{R}^n$ $|(x_1, x_2, \dots, x_n)| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$

Bilinear form:- Let V be a vector space over $\mathbb{R}$. A map $\beta: V \times V \rightarrow \mathbb{R}$ is called bilinear form if it satisfies

(i) $\beta(v_1 + v_2, w) = \beta(v_1, w) + \beta(v_2, w)$

(ii) $\beta(\alpha v, w) = \alpha \beta(v, w)$

(iii) $\beta(v, w_1 + w_2) = \beta(v, w_1) + \beta(v, w_2)$

(iv) $\beta(v, \alpha w) = \alpha \beta(v, w)$

$\left\{ \begin{array}{l} \forall v, w, v_1, v_2, w_1, w_2 \in V \\ \alpha \in \mathbb{R} \end{array} \right.$

• $V = \mathbb{R}^2$, $\beta: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$
 $\beta((x_1, y_1), (x_2, y_2)) = x_1 x_2 + y_1 y_2$

① $\beta((x_1, y_1) + (x'_1, y'_1), (x_2, y_2)) = \beta((x_1 + x'_1, y_1 + y'_1), (x_2, y_2))$
 $= (x_1 + x'_1) x_2 + (y_1 + y'_1) y_2$
 $= x_1 x_2 + y_1 y_2 + x'_1 x_2 + y'_1 y_2$
 $= \beta((x_1, y_1), (x_2, y_2)) + \beta((x'_1, y'_1), (x_2, y_2))$

② $\beta(\alpha(x_1, y_1), (x_2, y_2)) = \beta((\alpha x_1, \alpha y_1), (x_2, y_2))$
 $= \alpha x_1 x_2 + \alpha y_1 y_2 = \alpha(x_1 x_2 + y_1 y_2)$

$$= \alpha \beta((x_1, y_1), (x_2, y_2))$$

③ check! $\beta((x_1, y_1), (x_2, y_2) + (x'_2, y'_2)) = \beta((x_1, y_1), (x_2, y_2)) + \beta((x_1, y_1), (x'_2, y'_2))$

④ $\beta((x_1, y_1), \alpha(x_2, y_2)) = \alpha \beta((x_1, y_1), (x_2, y_2))$

* $\beta: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$

$$\beta((x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n)) = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

is a bilinear form.

- A Bilinear form $\beta: V \times V \rightarrow \mathbb{R}$ is called symmetric bilinear form if $\beta(v, w) = \beta(w, v) \quad \forall v, w \in V$.

Example

$$(x_1, y_1) \cdot (x_2, y_2) = x_1 x_2 + y_1 y_2$$

$$(x_2, y_2) \cdot (x_1, y_1) = x_2 x_1 + y_2 y_1$$

- A Bilinear form $\beta: V \times V \rightarrow \mathbb{R}$ is called positive definite Bilinear form if

$$\beta(v, v) \geq 0 \quad \forall v \in V$$

and $\beta(v, v) = 0 \quad \text{iff} \quad v = 0$.

Example

$$(x_1, y_1) \cdot (x_1, y_1) = x_1^2 + y_1^2 \geq 0$$

$$(x_1, y_1) \cdot (x_1, y_1) = x_1^2 + y_1^2 = 0 \quad \text{iff} \quad (x_1, y_1) = (0, 0)$$

- Dot product on $\mathbb{R}^n$ is a positive definite symmetric bilinear form.

Inner Product := positive definite symmetric bilinear form.

Inner Product :- Positive definite symmetric Bilinear form is called Inner product.

Inner Product space :- A vector space V is called Inner product space, if inner product is defined on it.

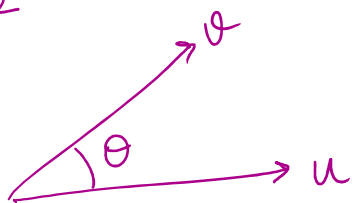
$(\mathbb{R}^n, \text{Dot product})$ is an Inner Product space.

on $\mathbb{R}^n$, $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$

$$\begin{aligned} |(x_1, x_2, \dots, x_n)| &= \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} \\ &= \sqrt{(x_1, x_2, \dots, x_n) \cdot (x_1, x_2, \dots, x_n)} \end{aligned}$$

• Let V be an inner product space. Then for any $v \in V$, $\|v\| = \sqrt{\beta(v, v)}$, where β is inner product defined on V .

• $\mathbb{R}^2$



$$u \cdot v = \|u\| \|v\| \cos \theta$$

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|}$$

• Let V be an inner product space with inner product β defined on it. Let u and v be two vectors in V , then angle θ b/w u and v is given by

two vectors in V , then $\cos \theta$ and θ is given by

$$\theta = \cos^{-1} \left(\frac{\beta(u, v)}{\sqrt{\beta(u, u)} \sqrt{\beta(v, v)}} \right)$$

- Two vectors u and v are s.t.b orthogonal to each other if $u \cdot v = 0$
- A vector $v \in V$ is called Normal vector if $\|v\| = 1$.

Example

$M_{3 \times 2}(\mathbb{R})$ forms a vector space $\mathbb{R}$

Claim: $\langle A, B \rangle = \text{trace}(A^t B)$ This is an inner product on $M_{3 \times 2}(\mathbb{R})$.

$$\begin{aligned} \langle A_1 + A_2, B \rangle &= \text{trace}((A_1 + A_2)^t B) \\ &= \text{tr}((A_1^t + A_2^t) B) \\ &= \text{tr}(A_1^t B + A_2^t B) \\ &= \text{tr}(A_1^t B) + \text{tr}(A_2^t B) = \langle A_1, B \rangle + \langle A_2, B \rangle. \end{aligned}$$

$$\begin{aligned} \langle \alpha A, B \rangle &= \text{trace}((\alpha A)^t B) \\ &= \text{tr}(\alpha (A^t B)) = \alpha \text{tr}(A^t B) = \alpha \langle A, B \rangle. \end{aligned}$$

$$\bullet \langle A, B_1 + B_2 \rangle = \langle A, B_1 \rangle + \langle A, B_2 \rangle \quad \bullet \langle A, \alpha B \rangle = \alpha \langle A, B \rangle.$$

• Check

$$\left. \begin{aligned} \langle A, B \rangle &= \text{tr}(A^t B) \\ \langle B, A \rangle &= \text{tr}(B^t A) \\ (B^t A)^t &= A^t B \\ \text{tr}(B^t A) &= \text{tr}(B^t A)^t = \text{tr}(A^t B) \end{aligned} \right\} \rightarrow \langle, \rangle \text{ is a symmetric Bilinear form}$$

• Check

$$\langle A, A \rangle \geq 0 \quad \& \quad \underline{\langle A, A \rangle = 0 \text{ iff } A = 0}$$

if $\underline{A \neq 0}$ then $\langle A, A \rangle > 0$

$$\boxed{\text{tr}(A^t A) > 0}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \quad A^t = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}$$

$$A^t A = \begin{bmatrix} a_{11}^2 + a_{21}^2 + a_{31}^2 & * \\ * & a_{12}^2 + a_{22}^2 + a_{32}^2 \end{bmatrix}$$

$$\text{tr}(A^t A) = a_{11}^2 + a_{21}^2 + a_{31}^2 + a_{12}^2 + a_{22}^2 + a_{32}^2 > 0$$

Orthogonal set:- Let V be an Inner product space over $\mathbb{R}$. A subset $S = \{v_1, v_2, \dots, v_k\}$ is called orthogonal set if $\langle v_i, v_j \rangle = 0$ $\forall i \neq j$.

orthonormal set.

$V = \mathbb{R}^2$, Inner product space

$S = \{(1, 0), (0, 1)\}$

$V = \mathbb{R}^2$, Inner product space

Example of orthogonal set

$$\{(1,0), (0,1)\}$$

Orthonormal set:- An orthogonal set $S = \{v_1, \dots, v_k\}$ is called orthonormal set if $\|v_i\| = 1 \forall i$.

• $\{(4,0), (0,5)\} \rightarrow$ orthogonal set
Not An orthonormal set.

* Let V be an inner product space. Let $S = \{v_1, v_2, \dots, v_k\}$ be an orthonormal set. Then S is a linearly independent set.

Solution:-

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0$$

T.P. $\alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_k = 0.$

$$\begin{aligned} &\rightarrow \|v_i\| = 1 \\ &\Rightarrow \langle v_i, v_i \rangle = 1 \end{aligned}$$

$$\begin{aligned} &\rightarrow \langle v_i, v_j \rangle = 0 \\ &\quad \forall i \neq j \end{aligned}$$

$$\langle \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k, v_1 \rangle$$

$$\begin{aligned} &= \alpha_1 \langle \underline{v_1}, v_1 \rangle + \alpha_2 \langle \underline{v_2}, v_1 \rangle + \dots + \alpha_k \langle \underline{v_k}, v_1 \rangle \\ &= \alpha_1 \end{aligned}$$

$$\langle 0, v_1 \rangle = \alpha_1$$

we need to prove $\langle 0, v \rangle = 0 \forall v \in V$

To conclude $\alpha_1 = 0$

$$\langle 0 | 0 \rangle = 0$$

$$\langle 0, v \rangle = \langle 0+0, v \rangle$$

$$\langle 0, v \rangle = \langle 0, v \rangle + \langle 0, v \rangle$$

$$\Rightarrow \langle 0, v \rangle = 0$$

$$\begin{aligned}
 \bullet \quad \langle 0, v_i \rangle &= \langle \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_i v_i + \dots + \alpha_r v_r, v_i \rangle \\
 &= \alpha_1 \underbrace{\langle v_1, v_i \rangle}_0 + \alpha_2 \underbrace{\langle v_2, v_i \rangle}_0 + \dots + \alpha_i \underbrace{\langle v_i, v_i \rangle}_1 + \dots + \alpha_r \underbrace{\langle v_r, v_i \rangle}_0 \\
 0 &= \alpha_i \quad \Rightarrow \quad S \text{ is a LI set.}
 \end{aligned}$$